

Lying to a Bayesian: Optimal Randomisation in Repeated Pricing

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Abstract

Reinforcement Learning solves a wide range of games (Chess, Go, Atari). Here we apply RL methods to a game-theoretic problem in economics : a repeated pricing game between a Bayesian-updating producer and a consumer with private willingness-to-pay. We established existence of Bayesian Nash Equilibrium for the discrete-action setting via Carbonell-Nicolau (2015), and of the mixed strategy Nash equilibrium for the continuous-action extension through Reny (1999). We have used three different approaches (Monte Carlo, Q-learning, Deep Q-Network) to simulate the benefits of randomisation for consumer's behaviour. All three approaches show a consistent result that $\epsilon^* < 1$, supporting randomisation as an obfuscation strategy. Tabular Q-learning shows no clear monotonic trend in $\epsilon^*(\beta)$, while DQN recovers the monotone decrease predicted by Monte Carlo, reconciling the previously-observed discrepancy. We empirically validate the Karp-Kleinberg lower bound on posterior variance, confirming that mixed strategies prevent posterior collapse.

1 Game Setting

Here I define a infinitely repeated game between consumer and producer.

- **Consumer**

Consumer has willingness to pay \$100.00 for a product, so his profit is the difference between his wtp and the price charged by the producer. Consumer's action space is $\{\pm 1\}$ where 1 represents take the producer's offer, while -1 represents reject the offer. Consumer's objective function is a sum of all expected future profits, i.e.

$$V_0(a) = \max_{a \in \{\pm 1\}} ((100 - x_0)\mathbb{1}(a = 1) + \beta E[V_1(a)])$$

the above is the consumer's bellman equation to be optimised where $\beta \in [0, 1]$ represents the time discount factor.

Consumer's policy :

$$\pi_t^c(a_t|x_t) = \epsilon \mathbb{1}(a_t = +1, x_t \leq 100) + (1 - \epsilon) \mathbb{1}(a_t \in \{\pm 1\})$$

where $\begin{cases} \pi_*(a_t|x_t) = \begin{cases} 1 & x_t \leq 100 \\ -1 & \text{otherwise} \end{cases} \\ u_t(a_t, x_t) = (100 - x_t) \mathbb{1}(a_t = +1) \end{cases}$. Here the baseline (deterministic) policy is

$$\bar{\pi}(a_t|x_t) = \mathbb{1}(a_t = +1) \mathbb{1}(x_t \leq 100) + \mathbb{1}(a_t = -1) \mathbb{1}(x_t > 100)$$

– Information hiding (Reputation Theory)

: if consumer always accepts when $p \leq 100$, and refuses when $p > 100$, producer perfectly infers $w^* = 100$ and always charges $x = 100$ forever, giving consumer 0 surplus. Randomisation creates ambiguity.

– Exploration-Exploitation (Bandit Theory)

: consumer trades off immediate surplus (exploiting good prices) against long-term surplus (maintaining uncertainty). The optimal ϵ^* trades these off, similar to the UCB's exploration bonus.

Then we have a consumer's value function

$$V(x_t) = E_{x_{t+1} \sim \mu, a_t \sim \pi_t(\cdot|x_t)}[u_t(a_t, x_t) + \beta V_{t+1}(x_{t+1}) | \pi_t, x_t \sim \pi_t(\cdot|x_t)]$$

• Producer

Producer produces a product at the cost of \$50.00. The producer observes consumer's behaviour given each price and updates their belief about consumer's willingness to pay. Then they aim to extract full profit by setting the price as the consumer's willingness to pay.

Producer assumes consumer's willingness to pay is drawn from an unknown distribution. By the Central Limit Theorem, its mean follows approximately normal distribution: $\mathcal{N}(\mu_t, \sigma_t^2)$. The producer charges $x_t = \mu_t$ (the posterior mean) and updates (μ_t, σ_t^2) via Bayesian updates after each observation.

– at $t \in \{0, \dots, 6\}$

the producer explores consumer's choice by playing "take a middle" policy.

– $t \geq 7$

the producer uses data to fit the normal distribution and get the mean and set the price as the updated mean.

Here I denote by μ a measure for the producer's distribution for the states (the state here means the price the producer sets).

Producer's policy :

$$\pi_t^p(x_t) = \begin{cases} \pi' & 0 \leq t < \lfloor \log_2(H - L) \rfloor \\ \mu_t & t \geq \lfloor \log_2(H - L) \rfloor \end{cases}$$

where μ_t is the mean of normal distribution updated from the past t -observations, and π' is the optimal policy for a "guess a number" game;

Algorithm 1 optimal policy π'

$t = 0$

while True **do**

For a interval $[L_t, U_t]$, $x_t = \lfloor (L_t + U_t)/2 \rfloor$

if $a_t = +1$ **then**

$L_{t+1} = x_t$

else

$U_{t+1} = x_t$

end if

if $L_{t+1} \neq U_{t+1}$ **then**

$t+ = 1$

else

break

end if

end while

Here I assume the range $[0,150]$, hence $\lfloor \log_2(H - L) \rfloor = 7$. Thus, for time $t \in \{0, \dots, 6\}$, producer explores consumer's behaviour by playing "take the middle" policy. At $t = 7$, the producer fits data into the normal distribution with mean μ_7 and standard deviation σ_7 , i.e. $\mu \sim N(\mu_7, \sigma_7^2)$. After $t = 7$, the producer updates his/her distribution by Bayesian updates.

2 Theoretical Foundations

2.1 Baseline Model

My baseline is that consumer plays truly to their preferences; if the price is higher than their willingness to pay, they reject it and conversely the price is lower than or equal to the willingness to pay, they accept it. Then it comes "guess-a-number" game to the producer, and they get maximum profits once they get the right number, while consumer only gets 0 surplus forever. In that game, the optimal policy is taking the middle number. In this setting, the producer sets price

$$\begin{aligned} (x_0, a_0) &= (75, +1) \\ (x_1, a_1) &= ((150 + 75)/2 \approx 112, -1) \\ (x_2, a_2) &= ((112 + 75)/2 \approx 93, +1) \\ \text{in range } [0, 150], \text{ the game will be } (x_3, a_3) &= ((93 + 112)/2 \approx 102, -1) \text{ and af-} \\ (x_4, a_4) &= ((102 + 93)/2 \approx 97, +1) \\ (x_5, a_5) &= ((102 + 97)/2 \approx 99, +1) \\ (x_6, a_6) &= ((102 + 99)/2 = 100, +1) \end{aligned}$$

ter trial 6, the producer's binary search converges to $w^* = 100$, and the producer will charge $p = 100$ forever from $t \geq 7$. In this case, the consumer's total profit is $25 + \beta^2 \cdot 7 + \beta^4 \cdot 3 + \beta^5 \cdot 1 + \beta^6 \cdot 0 + \sum_{i=7}^{\infty} \beta^i \cdot 0 = 25 + 7\beta^2 + 3\beta^4 + \beta^5$.

2.2 Noise Model

The base conjecture of this game is that randomisation obscures the signal so that the producer cannot observe the willingness to pay. This transforms the Baseline Model into a "guess with lies", and the producer side faces a noisy binary search problem.

Recall the deterministic baseline policy is $\bar{\pi}(a_t|x_t) = \mathbb{1}(a_t = +1)\mathbb{1}(x_t \leq 100) + \mathbb{1}(a_t = -1)\mathbb{1}(x_t > 100)$. A "lie" occurs when consumer deviates from it :

$\begin{cases} x_t \leq 100 \text{ but } a_t = -1 & \text{rejecting a good offer} \\ x_t > 100 \text{ but } a_t = +1 & \text{accepting a bad offer} \end{cases}$. Under such mixed policy, compute the conditional lie probabilities :

$\begin{cases} \mathbb{P}(a_t = -1|x_t \leq 100) = \epsilon \cdot 0 + (1 - \epsilon)(\frac{1}{2}) = \frac{1 - \epsilon}{2} \\ \mathbb{P}(a_t = +1|x_t > 100) = \epsilon \cdot 0 + (1 - \epsilon)(\frac{1}{2}) = \frac{1 - \epsilon}{2} \end{cases}$. Denote $\alpha = \mathbb{P}(x_t \leq 100)$. Then the lie rate is

$$\begin{aligned} p &= \alpha \mathbb{P}(a_t = -1|x_t \leq 100) + (1 - \alpha) \mathbb{P}(a_t = +1|x_t > 100) \\ &= \alpha \frac{1 - \epsilon}{2} + (1 - \alpha) \frac{1 - \epsilon}{2} = \frac{1 - \epsilon}{2} \end{aligned}$$

which implies the lie rate is independent of α , and therefore the noisy-search analysis becomes tractable.

Then based on noisy search theory [Karp2007], after t queries with lie rate p , posterior variance $\sigma_T^2 \geq C \exp(-TD(p))$ where $D(p) = (1/2 - p)^2$ is the discrimination rate. Then I have

$$\sigma_T^2 \geq C \exp(-T \cdot \epsilon^2/4) > 0$$

Therefore σ_T^2 is bounded below by a strictly positive constant depending on ϵ . This formally validates the strategy : randomisation prevents the producer's posterior from collapsing to a point mass at $w^* = 100$, ensuring positive expected consumer surplus over the infinite horizon. Explicitly, the consumer's value function is $V(x_t) = \mathbb{E}[(100 - x_t)\mathbb{1}(a_t = +1) + \beta \cdot V_{t+1}]$ and $\mathbb{P}(a_t = +1) = \epsilon \mathbb{1}(x_t \leq 100) + (1 - \epsilon)/2$. This optimisation problem can be solved iteratively by the tabular Q-learning algorithm.

3 Empirical Results

For further development, I have a bellman equation with continuous state (price) and discrete action (accept/reject). I took three approaches; Monte Carlo Simulation, Tabular Q-Learning and Deep Q-Network.

3.1 Phase 1

In phase 1 (the first 0-6 trials), the producer plays the "take the middle" policy like they are in the "guess a number" game, while consumer randomises its behaviour with ϵ . Then we have a randomised version of state variable and consumer's behaviour as

$$\begin{aligned} (x_0, a_0) &= (75, \frac{1+\epsilon}{2}) \\ (x_1, a_1) &= (\frac{149+75\epsilon}{2}, \frac{1-\epsilon^2}{2}) \\ (x_2, a_2) &= (74.5 + 37.5\epsilon - 19\epsilon^2, \frac{2+\epsilon+\epsilon^3}{4}) \\ (x_3, a_3) &= (\frac{594+336\epsilon-150\epsilon^2+36\epsilon^3}{8}, \frac{4+\epsilon-3\epsilon^2-\epsilon^3-\epsilon^4}{8}) \\ (x_4, a_4) &= (\frac{-9\epsilon^4+26\epsilon^3-180\epsilon^2+346\epsilon+593}{8}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\ (x_5, a_5) &= (\frac{5\epsilon^5-11\epsilon^4+68\epsilon^3-364\epsilon^2+703\epsilon+1183}{16}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \\ (x_6, a_6) &= (\frac{7\epsilon^5-7\epsilon^4+76\epsilon^3-368\epsilon^2+709\epsilon+1183}{16}, \frac{\epsilon^5+2\epsilon^4+4\epsilon^3-2\epsilon^2+3\epsilon+8}{16}) \end{aligned}$$

As a result, we reached somewhat a mixed-strategy equilibrium and the value for consumer at this phase becomes

$$\begin{aligned}
V(\epsilon, \beta) = & \frac{-\beta^6 \epsilon^5 + 45\beta^6 \epsilon^4 + 38\beta^6 \epsilon^3 - 462\beta^6 \epsilon^2 - 37\beta^6 \epsilon + 417\beta^6 + \beta^5 \epsilon^5 + 49\beta^5 \epsilon^4 + 46\beta^5 \epsilon^3}{32} + \\
& \frac{-250\beta^5 \epsilon^2 - 31\beta^5 \epsilon + 417\beta^5 + 4\beta^4 \epsilon^5 + 54\beta^4 \epsilon^4 + 64\beta^4 \epsilon^3 - 420\beta^4 \epsilon^2 - 20\beta^4 \epsilon + 414\beta^4 + 12\beta^3 \epsilon^4}{32} \\
& + \frac{8\beta^3 \epsilon^3 - 424\beta^3 \epsilon^2 - 8\beta^3 \epsilon + 412\beta^3 + 56\beta^2 \epsilon^3 - 296\beta^2 \epsilon^2}{32} \\
& + \frac{56\beta^2 \epsilon + 408\beta^2 - 408\beta \epsilon^2 + 408\beta + 400\epsilon + 400}{32}
\end{aligned}$$

For the closed form solution as a function of β , I have the first order condition;

$$\begin{aligned}
& -\frac{5}{32}\beta^6 + \frac{45}{8}\beta^6 \epsilon^3 + \frac{57}{16}\beta^6 \epsilon^2 - \frac{231}{8}\beta^6 \epsilon - \frac{37}{32}\beta^6 + \frac{5}{32}\beta^5 \epsilon^4 + \frac{49}{8}\beta^5 \epsilon^3 + \frac{69}{16}\beta^5 \epsilon^2 \\
& - \frac{225}{8}\beta^5 \epsilon - \frac{31}{32}\beta^5 + \frac{5}{8}\beta^4 \epsilon^4 + \frac{27}{4}\beta^4 \epsilon^3 + 6\beta^4 \epsilon^2 - \frac{105}{4}\beta^4 \epsilon - \frac{5}{8}\beta^4 \\
& + \frac{3}{2}\beta^3 \epsilon^2 - \frac{53}{2}\beta^3 \epsilon - \frac{1}{4}\beta^3 + \frac{21}{4}\beta^2 \epsilon^2 - \frac{37}{2}\beta^2 \epsilon + \frac{7}{4}\beta^2 - \frac{51}{2}\beta \epsilon + \frac{25}{2} = 0
\end{aligned}$$

but getting solution as a closed form requires unnecessary complexity. Instead, I plugged some different values of β in the above equation. As β differs from 0.5 to 0.99, ϵ^* varies from 0.57 to 0.07.

3.2 Phase 2

- **Monte Carlo**

I tabularised the continuous states (producer's distribution) and apply Monte-Carlo iteration to compute the value function and optimal ϵ for consumer's profits. For computation simplicity, I could not use the phase 1 data, instead assume $\mathcal{N}(100, 10^2)$. Then producer updates his belief about willingness to pay by Bayesian update with the observation of consumer's behaviour, and setting the new distribution's mean as a new price.

Here my pivotal argument is that if consumer is more forward-looking and therefore explore more, they earn more of surplus. It is derived by two results here. I first conjectured that myopic consumers (low β) prefer immediate optimal play over exploration, so ϵ should decrease with β . So I iterated with different time discount factor, and the result exactly shows the inverse correlation between them, the left of Figure 1. Also, when the consumer is more foresighted, in other words value more of future surplus, they get

Algorithm 2 Monte Carlo Simulation

$t, V = 0, 0$

while $t \leq 200$ **do**

$price = \text{mean}$

$r = \text{unif}(0, 1)$

if $r < \epsilon$ **then**

$a = +1$ **if** $price \leq wtp$ **else** $a = -1$

else

$a = \text{unif}(\{-1, +1\})$

end if

if $a = +1$ **then**

$V += \beta^t * (100 - price)$

end if

 update μ, σ via Bayesian rule given $(price, a)$

$t += 1$

end while

higher expected values, the right of Figure 1. These two empirical correlations confirm the theoretical hypothesis: patient consumers optimally trade immediate surplus for reputation, yielding higher long-run value.

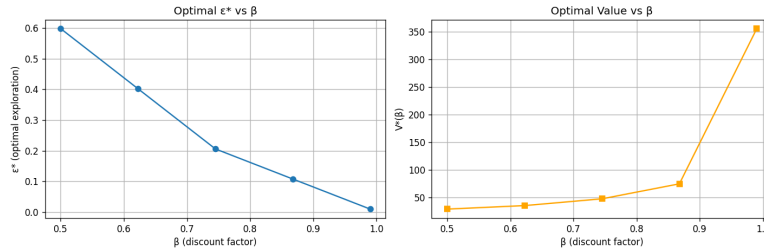


Figure 1: Correlations drawn from Monte Carlo

- **Tabular Q-Learning**

As set in section 1, producer's belief is $\mathcal{N}(\mu_t, \sigma_t^2)$ and producer charges $x_t = \mu_t$. As the name suggests, we discretise it onto the grid $A_0 = \{0, 1, \dots, 150\}$ for tabular Q -learning so that the Q -table is indexed by observed states. Then we can compute posterior probability $\mathbb{P}(wtp = x' | data_t) = \mathcal{N}(x'; \mu_t, \sigma_t^2)$. The consumer applies Q -learning to learn optimal ϵ given observed prices. The Q -table implicitly captures expected future surplus through the Bellman

backup. Firstly, define the Q-function for my problem

$$Q(x, \epsilon) = \mathbb{E}[u(x, \epsilon) + \sum_{j=1}^{\infty} \beta^j (100 - x_{t+j}) \mathbb{1}(a_{t+j} = +1) | x_t = x, \epsilon_t = \epsilon, \mu]$$

which has Bellman optimality equation

$$Q^*(x, \epsilon) = u(x, \epsilon) + \beta \sum_{x'} P_{xx'}(\epsilon) \max_{\epsilon'} Q^*(x', \epsilon'), \quad \forall x \in X, \epsilon \in \{0.0, 0.1, \dots, 1.0\}$$

With above equations, I can execute Q-learning update

$$Q_{t+1}(x_t, \epsilon_t) = (1 - \gamma) Q_t(x_t, \epsilon_t) + \gamma (u(x_t, \epsilon_t) + \beta \max_{\epsilon'} Q_t(x_{t+1}, \epsilon'))$$

where the learning rate is denoted as γ .

For state transition, I sample a fixed number of price states from the producer's posterior distribution, and build the Markov transition matrix based on that. As the game is repeated, this Markov transition matrix evolves.

CLAIM : There is a Bayesian Nash equilibrium which admits a stationary distribution.

Definition 3.1 (Carbonell-Nicolau, 2015). *A Bayesian game is a collection $\Gamma = ((T_i, \mathcal{T}_i), X_i, u_i, p)_{i=1}^N$ where $1, \dots, N$ is a finite set of players, $(T_i, \mathcal{T}_i)$ is a measurable space where T_i is player i 's type space, X_i is player i 's action space, $u_i : T \times X \rightarrow \mathbb{R}$ represents player i 's payoff function, and p is a probability measure on $(T, \otimes \mathcal{T}_i)$ denoting the common prior over type profiles.*

We define $i = 0$ (producer), $i = 1$ (consumer), with type spaces $T_0 = \{*\}$, $T_1 = \mathbb{R}_+$ (consumer's WTP), action spaces $A_0 = \{0, 1, \dots, 150\}$ (discretized prices), $A_1 = \{0, 0.1, \dots, 1.0\}$ (discretised exploration rates), and stage payoffs:

$$u_0(t, x, \epsilon) = \mathbb{E}_a[(x - 50) \mathbb{1}(a = +1) | x, \epsilon] = (x - 50) \cdot \mathbb{P}(a = +1 | x, \epsilon)$$

$$u_1(t, x, \epsilon) = \mathbb{E}_a[(t_1 - x) \mathbb{1}(a = +1) | x, \epsilon] = (t_1 - x) \cdot \mathbb{P}(a = +1 | x, \epsilon)$$

$a \sim \pi_\epsilon(\cdot | x)$ where $\mathbb{P}(a = +1 | x, \epsilon) = (\epsilon) \mathbb{1}(100 \geq x) + (1 - \epsilon)/2$ The common prior p marginalizes to $t_1 \sim \mathcal{N}(100, \sigma_w^2)$.

Condition : $\forall i, \forall \delta > 0, \exists$ a $(\mathcal{B}(X_i), \mathcal{B}(X_i))$ -measurable map $\phi : X_i \rightarrow X_i$ s.t. the following holds ; $\forall (t, x) \in T \times X, \exists V_{x_{-i}}$ of x_{-i} s.t.

$$u_i(t, (\phi(x_i), y_{-i})) > u_i(t, (x_i, x_{-i})) - \delta, \quad \forall y_{-i} \in V_{x_{-i}}$$

Proposition 3.1 (Carbonell-Nicolau, 2015). *If the Bayesian game satisfies the above condition, it is uniformly payoff secure.*

Verification of Condition (Proposition 3.1):

Both A_0 and A_1 are finite with discrete topology; for any action a , $\{a\}$ is open. Take $\phi_i = \text{id}$ for both players.

For producer ($i = 0$): $\forall(t, x, \epsilon)$, take $V_\epsilon = \{\epsilon\}$. Then

$$u_0(t, (\phi_0(x), \epsilon)) - u_0(t, (x, \epsilon)) = 0 > -\delta. \quad \checkmark$$

For consumer ($i = 1$): $\forall(t, x, \epsilon)$, take $V_x = \{x\}$. Then

$$u_1(t, (x, \phi_1(\epsilon))) - u_1(t, (x, \epsilon)) = 0 > -\delta. \quad \checkmark$$

By Proposition 3.1, Γ is uniformly payoff secure.

USC of sum: Since u_0 and u_1 are continuous on the discrete action space, both are USC. Their sum is USC.

Absolute continuity of prior: Since $T_0 = \{*\}$ is trivial, $p(t_0, t_1) = \delta_*(t_0) \cdot p_1(t_1)$, absolutely continuous w.r.t. the product of marginals.

Theorem 3.1 (Carbonell-Nicolau, 2015). *Suppose the Bayesian game $\Gamma = ((T_i, \mathcal{T}_i), X_i, u_i, p)_{i=1}^N$ is uniformly payoff secure, and $\forall t \in T$, $\sum_{i=1}^N u_i(t, \cdot) : X \rightarrow \mathbb{R}$ is upper semi continuous. If p is absolutely continuous w.r.t. $p_1 \otimes \dots \otimes p_N$, then there exists a Bayes-Nash equilibrium of Γ .*

Then by Theorem 3.1, this game has a Bayesian Nash Equilibrium.

We conjecture that stationary play of the stage BNE constitutes a Markov perfect equilibrium of the discounted repeated game, with stationary joint distribution over $(\mu_t, \sigma_t^2, \epsilon_t)$. For the empirical validation, I simulate the game below. So even though the price state is not exactly Markov in the absence of producer's belief, we conjectured convergence under the BNE-induced stationary distribution.

With the exploration $v \neq 0$, and stationary state distribution (BNE conjecture), Q-learning algorithm empirically converges and therefore the above algorithm will give approximately true optimal policy. Also, here we only have the information about cumulative distribution function, not the point estimate, and therefore we use cdf to update the normal distribution (exact python code is provided at the github). The result gives no clear monotone

Algorithm 3 Tabular Q-Learning

Input: β time discount factor, initial normal distribution p_0 , η convergence constant, T_{max} maximum iteration, learning rate γ , exploration rate ν .
Initialize Q_0 with zeros, $x_0 = \text{mean}(p_0)$, $t_0 = 0$.
while $t < T_{max}$ **do**
 Observe x_t and find $\epsilon_t^* = \arg\max Q_t(x, \epsilon)$
 With probability ν , play randomly from epsilon grid. With probability $1 - \nu$, play ϵ_t^* .
 Sample $a_t \sim \pi_{\epsilon^*}(\cdot|x_t)$, where $\pi_{\epsilon^*}(a = +1|x_t) = \epsilon^* \mathbb{1}(x \leq 100) + (1 - \epsilon^*)/2$
 Observe $u_t(x_t, \epsilon_t^*) = (100 - x_t) \mathbb{1}(a_t = +1)$.
 update $\begin{cases} p_t \text{ with Bayesian update given } (x_t, a_t) \\ x_{t+1} = \text{mean}(p_{t+1}) \end{cases}$
 $Q_{t+1}(x_t, \epsilon_t^*) = Q_t(x_t, \epsilon_t^*) + \gamma(u_t(x_t, \epsilon_t^*) + \beta \max_{\epsilon'} Q_t(x_{t+1}, \epsilon') - Q_t(x_t, \epsilon_t^*))$
 if $\|Q_{t+1} - Q_t\|_\infty < \eta$ **then**
 Break
 end if
 $t = t + 1$
end while
return Q, π^*

correlation between time discount factor and optimal epsilon value. The optimal epsilon hovers around 0.4 to 0.6 across time discount factor β , with no clear monotone dependence. We later show in the subsequent section that this is an artifact of state discretisation; DQN with continuous state input recovers the monotone trend predicted by Monte Carlo. Posterior σ remains bounded from 0 across all β , consistent with the noisy-search lower bound.

- **Deep Q-Learning**

Here we also use the same setting with slight adjustments, we define the game. We define $i = 0$ (producer), $i = 1$ (consumer), with action spaces $A_0 = [0, 150]$ (continuous price), $A_1 = [0, 1]$ (continuous exploration rates), and stage payoffs:

$$\begin{aligned} u_0(x, \epsilon) &= (x - 50) \cdot \mathbb{P}(a = +1|x, \epsilon) \\ u_1(x, \epsilon) &= (100 - x) \cdot \mathbb{P}(a = +1|x, \epsilon) \end{aligned}$$

$a \sim \pi_\epsilon(\cdot|x)$ where $\mathbb{P}(a = +1|x, \epsilon) = (\epsilon) \mathbb{1}(100 \geq x) + (1 - \epsilon)/2$. Here the model no longer requires a Bayesian-game framework with types, we apply Reny's existence theorem[Reny1999] for discontinuous games rather than Carbonell-Nicolau.

Proposition 3.2 (Reny,1999). *If $G = (X_i, u_i)_{i=1}^N$ is reciprocally u.s.c. and payoff secure, then it is better-reply secure.*

Theorem 3.2 (Reny,1999). *If $G = (X_i, u_i)_{i=1}^N$ is compact, quasiconcave, and better-reply secure, then it possesses a pure strategy Nash equilibrium.*

For each i , let M_i denote the set of Borel probability measures on A_i . The mixed extension $\tilde{G} = (M_i, \bar{u}_i)_{i=1}^N$ has payoffs $\bar{u}_i(\mu_1, \dots, \mu_N) = \int_{A_1 \times \dots \times A_N} u_i(x_1, \dots, x_N) d(\mu_1 \otimes \dots \otimes \mu_N)$.

Corollary 3.1 (Reny,1999). *Suppose $G = (X_i, u_i)_{i=1}^N$ is a compact Hausdorff game. Then G possesses a mixed strategy Nash equilibrium if its mixed extension $\tilde{G}$ is better reply secure.*

Verification of Condition (Proposition 3.2): Here we define

$$\Phi_0(x) = \begin{cases} x & \text{if } x \in [0, 100 - \nu] \cup [100 + \nu, 150] \\ 100 & \text{if } x \in [100 - \nu, 100 + \nu] \end{cases}$$

where ν is arbitrarily small number > 0 for producer and

$$\Phi_1(\epsilon) = \epsilon$$

for consumer.

– producer

for $i = 0$, $u_i(x_i^*, a'_{-i}) > u_i(x_i^*, a_{-i}) - \delta$
 $\Leftrightarrow u_0(x^*, \epsilon_1) - u_0(x^*, \epsilon_0) < \delta$
 $\Leftrightarrow (x^* - 50) \cdot (\mathbb{P}(a = +1|x = x^*, \epsilon_1) - \mathbb{P}(a = +1|x = x^*, \epsilon_0)) < \delta$
 $\Leftrightarrow (x^* - 50) \cdot [(\epsilon_1 - \epsilon_0)\mathbb{1}(x^* \leq 100) - \frac{1}{2}(\epsilon_1 - \epsilon_0)] < \delta$ hence
 $|u_0(x^*, \epsilon_1) - u_0(x^*, \epsilon_0)| \leq |K| \cdot |\epsilon_1 - \epsilon_0|$ where $K = (x^* - 50)[\mathbb{1}(x^* \leq 100) - 1/2]$
(here if $K = 0$, $u_0 = 0$, trivial) . Taking $V_{\epsilon_0} = \epsilon_1 : |\epsilon_1 - \epsilon_0| < \delta/|K|$ (or $V_{\epsilon_0} = A_1$ if $K = 0$) gives payoff security at x^* .

But here we have a discontinuity at $x = 100$, which should be dealt separately. This means when $x^* = 100$, there is a jump in ϵ since $\mathbb{P}(a = +1|x = 100) = \frac{1+\epsilon}{2}$ while $\mathbb{P}(a = +1|x = 100 + \eta) = \frac{1-\epsilon}{2}$, $\forall \eta > 0$. So at $x = 100 + \eta$, where η is sufficiently small so that $x \in [100 - \nu, 100 + \nu]$. Then

for a fixed ϵ_1 with varying ϵ_0 $\left\{ \begin{array}{l} \text{LHS :} \\ \text{RHS :} \end{array} \right. \begin{array}{l} u_0(100, \epsilon_0) = 50 \frac{1+\epsilon_0}{2} \\ u_0(100 + \eta, \epsilon_1) - \delta = \frac{1-\epsilon_1}{2}(50 + \eta) - \delta \end{array}$.

When $\epsilon_0 = \epsilon_1$, $25(1 + \epsilon_1) \geq 25(1 - \epsilon_1) + \eta(1 - \epsilon_1)/2 - \delta \Leftrightarrow 50\epsilon_1 + \delta \geq \eta(1 - \epsilon_1)/2$, $\forall \epsilon_1 > 0$ with sufficiently small $\eta > 0$. By the continuity in ϵ , payoff secure follows.

– consumer

for $i = 1, u_1(x, \epsilon) = (100 - x) \cdot \mathbb{P}(a = +1 | x, \epsilon)$ is jointly continuous on $A_0 \times A_1$ (since $(100 - x)$ vanishes at $x = 100$, discontinuity in $\mathbb{P}$ is eliminated). Thus payoff security holds with $\Phi_1(\epsilon) = \epsilon$.

Reciprocal USC of sum : $u_0(\cdot) + u_1(\cdot)$ is continuous except at $x = 100$. Therefore, it is u.s.c. for those continuous intervals. At $x = 100$,

$$\begin{cases} u_0(100, \epsilon) + u_1(100, \epsilon) = 50 \frac{1+\epsilon}{2} + 0 = 25(1 + \epsilon) \\ \lim_{x \rightarrow 100^+} u_0 + u_1 = 50 \frac{1-\epsilon}{2} + 0 = 25(1 - \epsilon) \\ \lim_{x \rightarrow 100^-} u_0 + u_1 = 50 \frac{1+\epsilon}{2} + 0 = 25(1 + \epsilon) \end{cases} .$$

Therefore $\limsup_{x \rightarrow 100} (u_0 + u_1) = 25(1 + \epsilon) = (u_0 + u_1)(100, \epsilon)$, so the sum is upper semicontinuous at $x = 100$. Elsewhere $u_0 + u_1$ is continuous, hence USC on $A_0 \times A_1$.

Then by Proposition 3.2., it is better-reply secure, and by Corollary 3.1. it has the mixed strategy Nash Equilibrium.

As set in the previous section, producer’s belief is $\mathcal{N}(\mu_t, \sigma_t^2)$ and producer charges $x_t = \mu_t$. This is known to the consumer, so consumer observes x_t and computes μ_t, σ_t based on the previous data. This is similar setting to the original DQN paper[Mnih2015], where they observe raw pixels and process them into another state variable. For the DQN implementation, we discretise the consumer’s action space A_1 to a finite grid $\{0, 0.01, 0.02, \dots, 1\}$ of 101 values. Existence of mixed-strategy NE for the continuous game suggests the discretised game has approximate NE for sufficiently fine grids. Additionally, we adopt the standard ϵ -greedy exploration scheme; to avoid the notation conflict with the consumer’s mixing parameter ϵ , we denote the exploration probability by ν throughout.

After executing simulations with 50000 iteration under the multiseed condition, we found out the result as in the table 1. This result shows a monotonic decreasing trend in ϵ^* as a function of the time discount factor β , consistent with the Monte Carlo finding. Across all β values, $\epsilon^* < 1$ confirming the central hypothesis that randomisation strictly improves consumer surplus over the deterministic policy. The table also reveals high variance at $\beta = 0.9$, which means our ϵ^* can be unstable at intermediate β , indicating the Q-function is relatively flat over ϵ in the regime, so that the argmax is sensitive to seed-level noise.

Algorithm 4 Deep Q-Network

Input: β (discount), $T_{\max}$ (max iterations), γ (learning rate), ν (exploration rate), η (minibatch size), K (target update period), $|\mathcal{D}|$ (buffer capacity), initial prior p_0 .

Initialize Q_Θ with random weights; target network Q_{Θ^-} with $\Theta^- \leftarrow \Theta$.

Initialize replay buffer $\mathcal{D}$ with capacity $|\mathcal{D}|$.

Initialize $s_0 = (\mu_0, \sigma_0)$ from p_0 ; $t \leftarrow 0$.

while $t < T_{\max}$ **do**

 Producer posts $x_t = \mu_t$.

$\epsilon_t^* = \arg\max_{\epsilon} Q_\Theta(s_t, \epsilon)$.

 With probability ν , sample ϵ_t uniformly from the action grid; with probability $1 - \nu$, $\epsilon_t = \epsilon_t^*$.

 Sample $a_t \sim \pi_{\epsilon_t}(\cdot | x_t)$ and observe $u_t = (100 - x_t) \cdot \mathbb{1}(a_t = +1)$.

 Producer's Bayesian update: $p_{t+1} \sim \mathcal{N}(\mu_{t+1}, \sigma_{t+1}^2)$ given (x_t, a_t) .

$s_{t+1} = (\mu_{t+1}, \sigma_{t+1})$; store $(s_t, \epsilon_t, u_t, s_{t+1})$ in $\mathcal{D}$.

if $|\mathcal{D}| \geq \eta$ **then**

 Sample random minibatch $\{(s_j, \epsilon_j, u_j, s'_j)\}_{j=1}^\eta$ from $\mathcal{D}$.

$y_j = u_j + \beta \cdot \max_{\epsilon'} Q_{\Theta^-}(s'_j, \epsilon')$.

$\Theta \leftarrow \Theta - \gamma \cdot \nabla_{\Theta} \left[\frac{1}{\eta} \sum_j (y_j - Q_\Theta(s_j, \epsilon_j))^2 \right]$.

end if

if $t \bmod K = 0$ **then**

$\Theta^- \leftarrow \Theta$.

end if

$t \leftarrow t + 1$.

end while

Return Q_Θ .

Table 1: DQN-learned ϵ^* across discount factors β (5 seeds each, $T = 50000$).

β	ϵ^* (mean)	std
0.50	0.558	0.233
0.70	0.520	0.247
0.90	0.252	0.273
0.99	0.118	0.050

4 Conclusion

All empirical results coincide in the sense that they all show the benefits of randomisation: optimal $\epsilon^* > 0$, indicating randomisation improves on deterministic play in the consumer-producer game. The producer’s posterior variance σ remains bounded above zero in all simulations, empirically validating the noisy-search lower bound.

The methods disagree on the precise function form of $\epsilon^*(\beta)$. By construction, Monte-Carlo grid-searches over fixed ϵ , while Q-learning was intended to learn a state-adaptive ϵ^* . However, tabular Q-learning’s coarse 1D discretisation of the 2D belief state (μ, σ) collapses this capability. DQN’s neural function approximator handles the continuous 2D state directly, recovering the monotone $\epsilon^*(\beta)$ trend predicted by Monte Carlo. This explains the empirical reconciliation.

This convergence of three independent methods (Monte Carlo, tabular Q-learning, and DQN) on the qualitative finding $\epsilon^* < 1$ with the quantitative agreement between MC and DQN, provides strong empirical support for randomisation as an optimal strategy in the obfuscation game.

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